

Interpreting a Box Plot

While the median tells you about the center of a data set, you can use a **box plot** to answer questions about an entire set of data.

To create a box plot, you need five numbers: the **minimum** (the lowest value in the set), the **lower quartile**, the median, the **upper quartile**, and the **maximum** (the highest value in the set). The lower quartile is the median of the lower half of the data. The upper quartile is the median of the upper half.

Example

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Ms. Rivera recorded the following test scores. Find the upper and lower quartiles of the data.

55 63 68 76 79 80 80 81 83 93 94 98

Step 1

Find the median. Make sure the data is arranged in order from least to greatest.
Of the 12 values, the two middle values are 80. The median is 80.

Step 2

Split the data in half at the median. Find the median of the lower half.

55 63 68 76 79 80

$68 + 76 = 144$
 $144 \div 2 = 72$

The median of the lower half is 72.

Step 3

Find the median of the upper half.

80 81 83 93 94 98

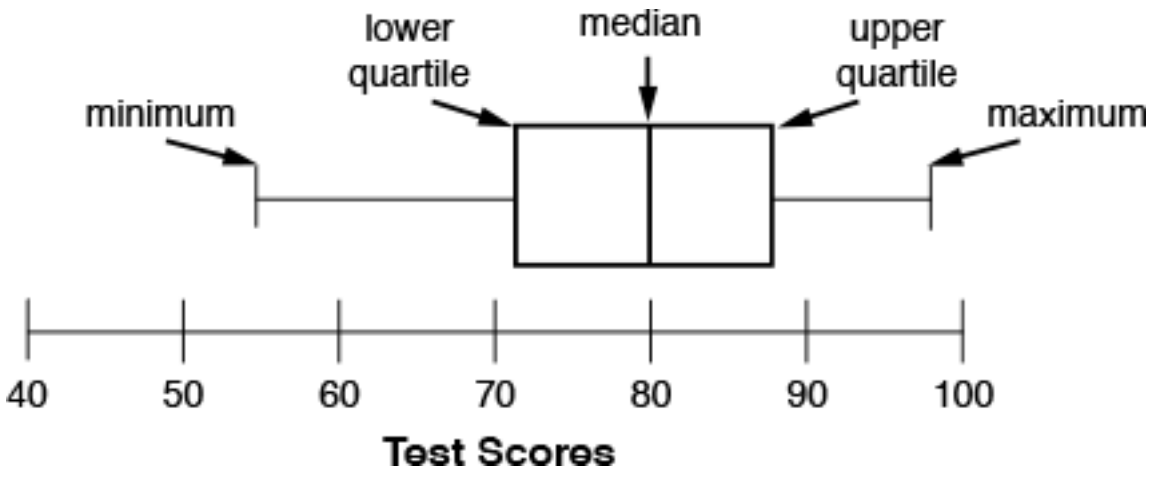
$83 + 93 = 176$
 $176 \div 2 = 88$

The median of the upper half is 88.

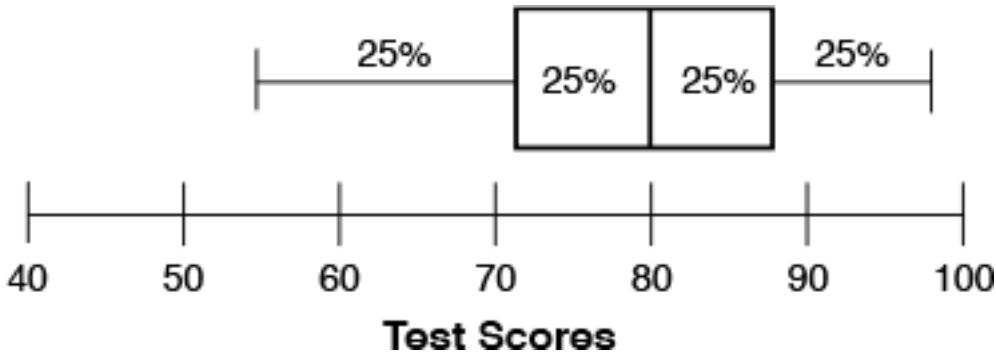
The lower quartile is **72**. The upper quartile is **88**.

A box plot is drawn on a scale. Draw a box that starts at the lower quartile and ends at the upper quartile. Draw a vertical line inside the box at the median of the entire data set. Then draw lines extending to the minimum and the maximum. These lines are called **whiskers**. This kind of chart is also called a **box-and-whisker plot**.

Look at the box plot of the test scores.



Each section of the plot represents 25% of the test scores.



Students in the *bottom* 25% of Ms. Rivera’s class scored between 55 and 72. Students in the *top half* of her class scored between 80 and 98.

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